Utility, Demand, Supply and Equilibrium

Week 8
Chapter 3

Utility, Consumption and Consumers

- ✓ **Utility:** In economics, the capability of any good to satisfy people's needs is termed as utility.

- ✓ **Consumption:** By using the goods we can just get their utility. We have to keep in mind that if the utility of any good is destroyed in any way other than satisfying a need, it will not be considered as consumption. That is, in economics, to finish the utility of any good for the purpose of satisfying a want or need, is known as consumption.

- ✓ **Consumer:** Other than the free accessibly goods, the person who is prepared to spend money to consume any other good, she/he is known as the consumer.

Cateris Paribus

Other-things-equal assumption (Ceteris Paribus)—the assumption that factors other than those being considered do not change. They assume that all variables except those under immediate consideration are held constant for a particular analysis.

For example, consider the relationship between the price of Pepsi and the amount of it purchased. Assume that of all the factors that might influence the amount of Pepsi purchased (for example, the price of Pepsi, the price of Coca-Cola, and consumer incomes and preferences), only the price of Pepsi varies. This is helpful because the economist can then focus on the relationship between the price of Pepsi and purchases of Pepsi in isolation without being confused by changes in other variables.

Total Utility and Marginal Utility

Total Utility

At the market, you want to buy more than one mango to eat. The money that you spend to buy the first mango, to buy for the second or third or fourth time, you do not spend the same. Because, after consuming the first mango, your wish to eat mango is satisfied to quite an extent. Your wish to eat mango for the second time, is lessened. For the third or fourth mango it will be even lesser. It might even be possible that you will not buy any more mangoes. Because, you do not have the wish to eat mango any longer, in other words, the utility of extra mangoes to you is zero.

The sum of the satisfaction from different units of a good derived at a particular time is known as the total utility. Since the extra mangoes gradually give lesser satisfaction, therefore, with the increment of consumption, the total utility increases at a decreasing rate.

Marginal Utility:

Suppose, you have bought 3 mangoes. Now you buy another mango. This fourth extra mango is the marginal mango. From this marginal mango, the satisfaction or the utility you derive is the marginal utility. That is, consuming one extra unit of good or service, the utility that one can get is marginal utility

Presentation of Total Utility and Marginal Utility by a List

Unit of Good	Total Utility (in taka)	Marginal Utility (in taka)
1st	8	8.00
2nd	8+7=15	7.00
3rd	15+6=21	6.00
4th	21+5=26	5.00
5th	26+4=30	4.00
6th	30+0=30	0.00
7th	30-1=29	-1.00

It can be seen in the list above that when the price of the 1st mango is tk 8 then the marginal utility is tk 8. Buying the 2nd mango makes the total cost tk 15. The total utility derived from 2 mangoes is tk 15. And thus the marginal utility, chronologically is, 8, 7, 6, 5, 4, 0, and -1 tk. And you will not buy the 7th mango since it has no utility to you. That means to consume the same product over and over again causes the marginal utility to diminish accordingly. This tendency of the marginal utility to fall due to the increment in the unit of consumption is known as the diminishing law of marginal utility.

Calculation of Marginal utility and average utility

❑ Marginal Utility

$$X = \frac{Changes\ in\ total\ utility}{Changes\ in\ number\ of\ Units}$$

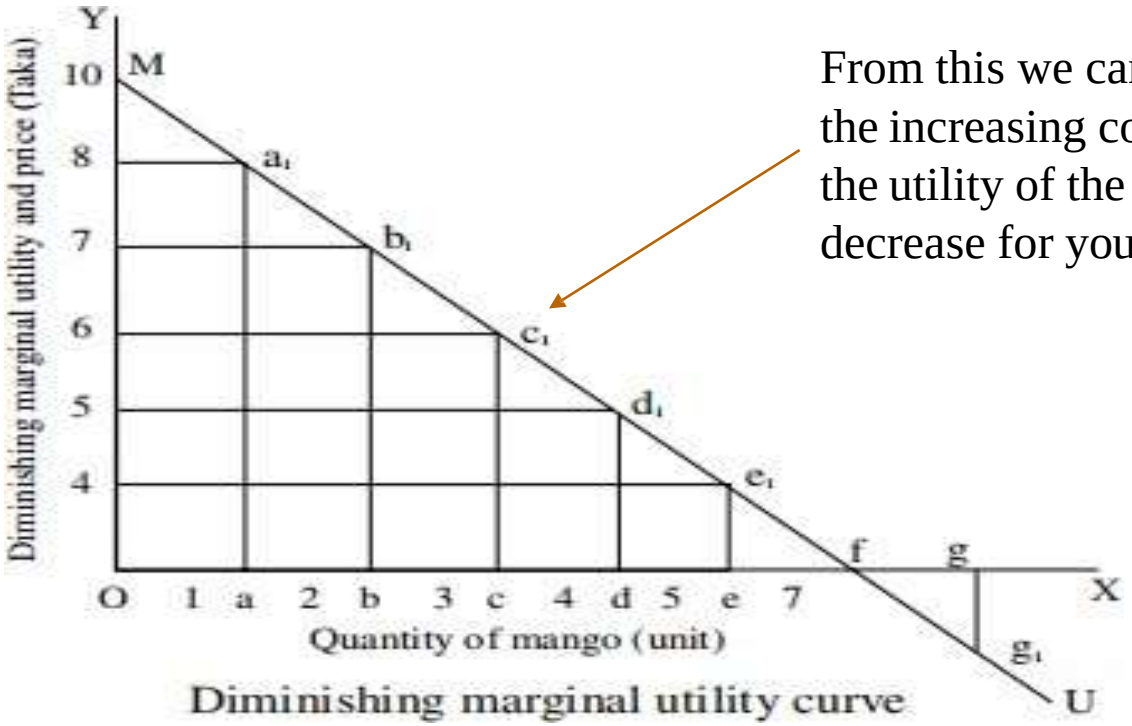
❑ Average Utility

$$X = \frac{total\ utility}{number\ of\ Units}$$

The Diminishing Law of Marginal Utility

The diminishing law of marginal utility maintains some conditions. These are,

a) the consumer should have general capacity of judgement;
b) the consumer can measure the utility of the good in terms of money if she/he wishes to;
c) the price of the good is to be equal to the marginal utility;
d) there will be no change in the income, taste and liking of the consumer while consuming the good.



Diminishing marginal utility curve

From this we can understand that with the increasing consumption of mangoes the utility of the extra units gradually decrease for you.

Demand

Three conditions are to be fulfilled in to make a demand in economics. e.g.-

1. the wish to get any good,
2. more than necessary capacity to buy, and
3. the wish to spend money to get that good.

Therefore, if a buyer has the wish to buy a good, the capacity to buy it and the wish to buy it at fixed price, then it is called a demand in economics. All else being constant or unchanged, at a particular time, if the price of the good falls the quantity of demand rises and if the price rises, the quantity demanded falls.

Demand Schedule or Demand List:

Price of each unit of good (in taka)	Quantity of demand (unit)
8.00	4
6.00	8
4.00	12
2.00	16

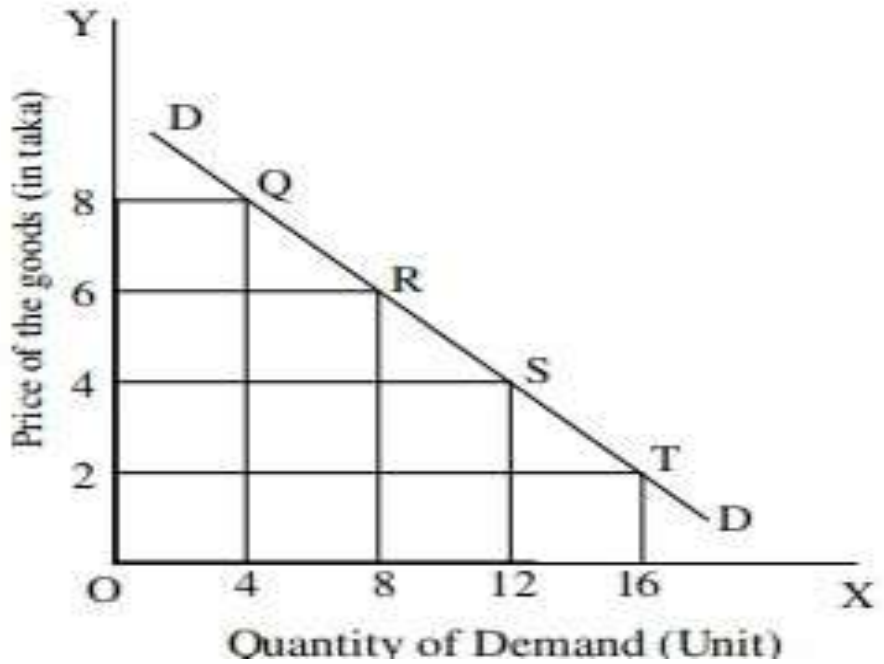
Market Demand: is an accumulation of all individual's demand at different prices.

Change in Demand: change in one or more of the determinants of demand will change the demand.

Demand Curve

A personal demand curve can be drawn from a person's demand schedule. Similarly, it is also possible to draw a market demand curve. The sum total of the personal or different demands of all the consumers at a particular price in the market is known to be the market demand.

In the diagram above, the horizontal axis or the OX axis shows the quantity of demand and the vertical axis or OY axis shows the price of the good. When the price of the good is tk 8 then the quantity of demand is 4 units and the two lines join at the point Q. Likewise, 8 units at tk 6, 12 units at tk 4 and 16 units at tk 2 is shown at the points R, S and T. Now, if we join the points Q, R, S, and T, we will get the DD line. This DD line is the demand curve. The points on the DD line are indicating the different quantities of demand at different prices of a good.



Week 9

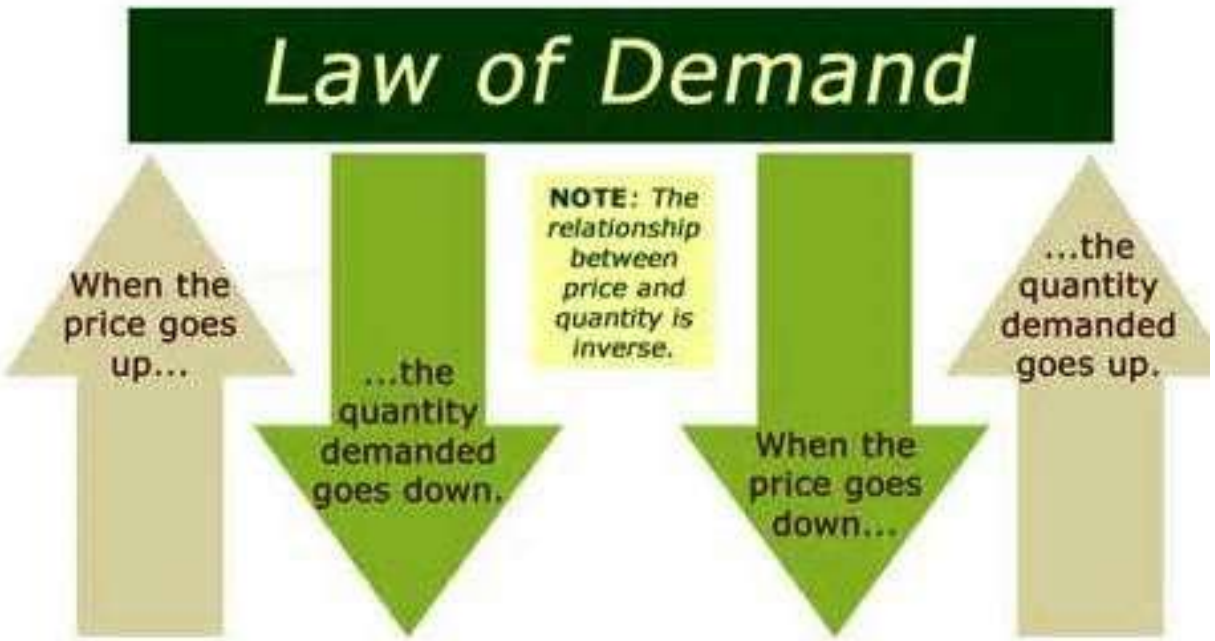
Utility, Demand, Supply and Equilibrium

Chapter 3

The Law of Demand

The law of demand states that, if all other factors remain equal, the higher the price of a good, the less people will demand that good.

In other words, the higher the price, the lower the quantity demanded.

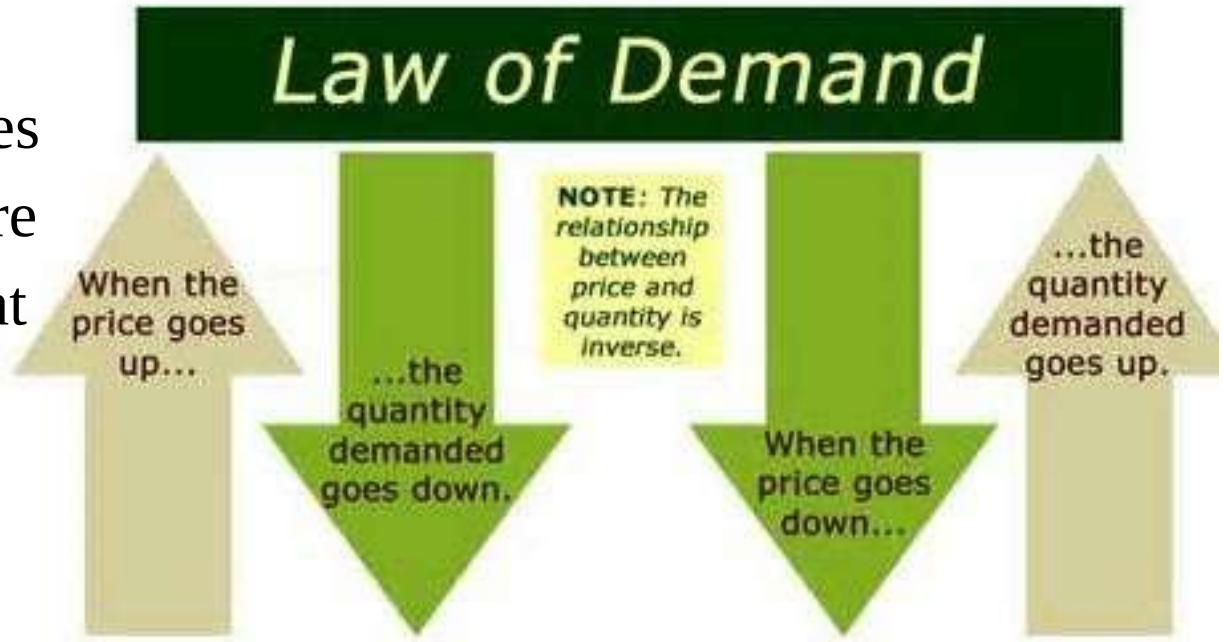


Why the inverse relationship between price and quantity demanded?

Diminishing marginal utility- each successive unit of the product consumed, consumer receives less benefit. Therefore, consumers will buy more units of commodities only when the price of that product begin s to fall.

Income effect- indicates that a lower price increases the purchasing power of a buyer's money income

Substitution effect- suggests that at lower price buyers have the incentive to substitute what is now a less expensive product for similar products that are now relatively more expensive. Example, Chicken and Beef.



Determinants of demand: Factors that shift the demand curve and cause more or less to be bought at any possible price. Includes the followings,

- ✓ Tastes- favorable change in preference will increase demand for a product and unfavorable change will reduce the demand.
- ✓ Number of buyers- An increase in the number of buyers in a market is likely to increase product demand.
- ✓ Income- For most products, a rise in income causes an increase in demand
  - -Normal goods: Products whose demand varies directly with money income are called superior goods, or normal goods. Examples, Furniture, Electronic Equipment.
  - -Inferior goods: Goods whose demand varies inversely with money income are called inferior goods. Example, used car, discount clothing.
- ✓ Price of related goods
  - -Substitute goods: one that can be used in place of another good. Example, Tea vs Coffee.
  - -Complementary goods: one that is used together with another good. Example, Car and Fuel.
  - -Unrelated goods: Goods that are not related to each other. Example, Butter and Golf ball.

Consumer expectations- consumer expectation about price increase or stock out will increase the demand for the goods.

Government policies and regulations: Government intervention can have a significant effect on supply.
Government intervention can take many forms including environmental and health regulations, hour and wage laws, taxes, electrical and natural gas rates and zoning and land use regulations.

Other factors--

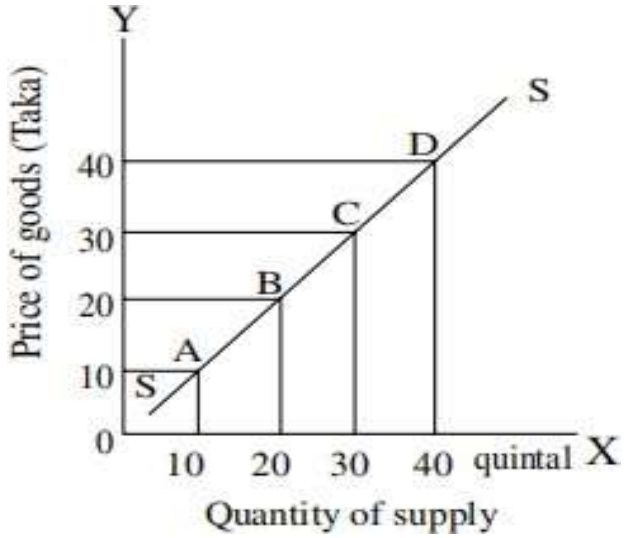
1. Tax and subsidy
2. Weather
3. Joint product (raw hide and beef)

Supply Curve

In economics, the quantity of a good that a producer is willing and capable to supply at a specified time and at a specific price, is known as supply. To mention, one good, a specified time and a specific price is considerable here. Therefore, the quantity of a product that a producer is willing to supply at a price is known to be supply in economic terms.

The Supply Schedule

Price of each unit of the product (in taka)	Quantity of Supply (quintal)
10.00	10
20.00	20
30.00	30
40.00	40



Market Supply: Market supply is derived from individual supply in exactly the same way that market demand is derived from individual demand.

Law of Supply: Sates that , all other things remain constant, the quantity supplied will increase as prices rise and the quantity supplied will decrease as prices fall.

Determinants of Supply: the basic determinants of supply are,

Resource prices- Higher resource prices increase production costs which decrease profit and supply. Example, high prices of sand and crushed rock reduces concrete supply.

Technology- Improvements in technology (techniques of production) enable firms to produce units of output with fewer resources.

Taxes and Subsidies- Increase in sales or property taxes will increase production costs and reduce supply.

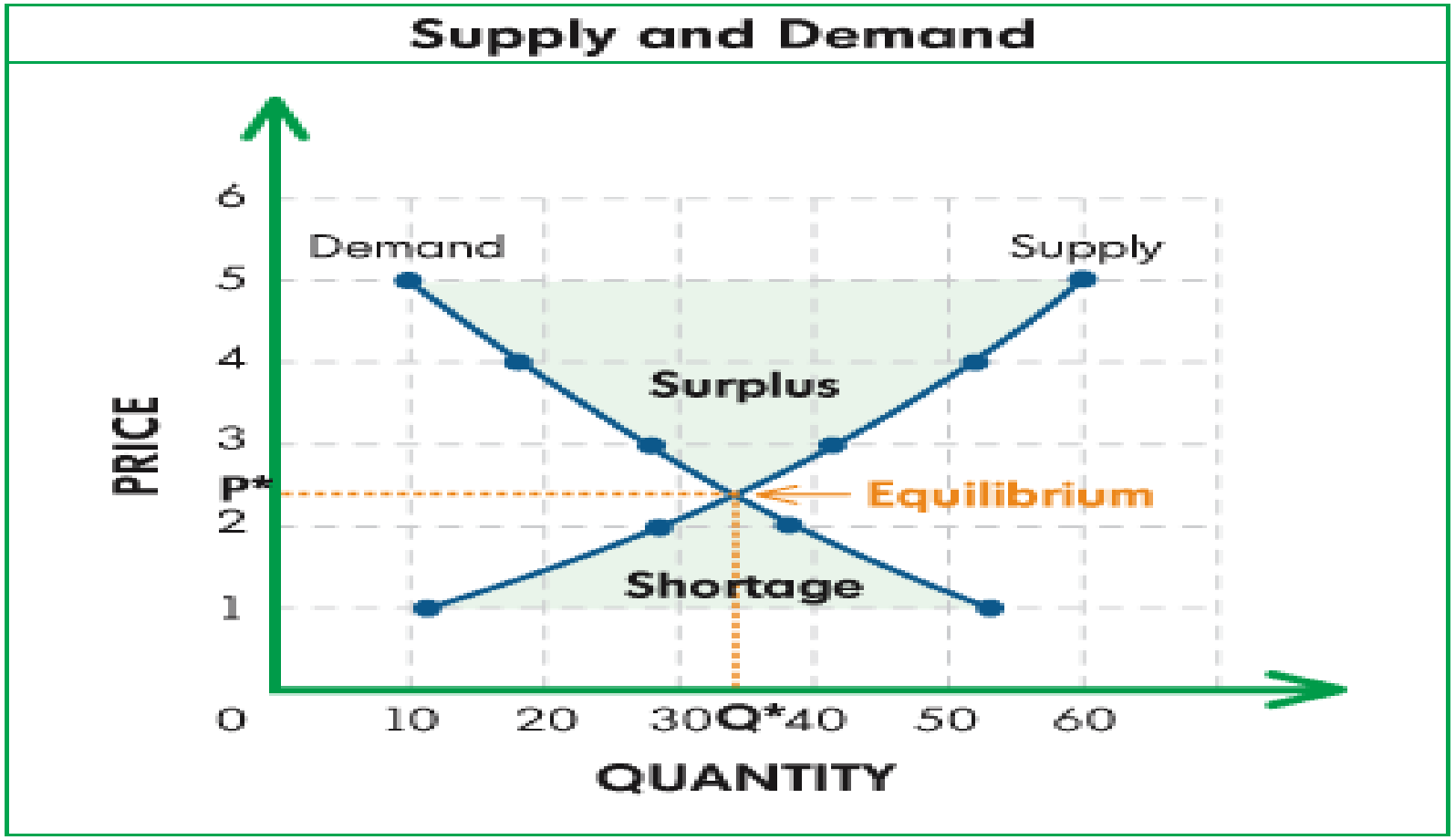
Prices of Other Goods- low prices of substitute products decline the supply of original product. Example, Soccer ball vs Volleball.

Producer Expectations- Changes in expectations about the future price of a product may affect the producer's current willingness to supply that product. Example, Farmers anticipating a higher wheat price in the future might withhold some of their current wheat harvest from the market, thereby causing a decrease in the current supply of wheat.

Number of Sellers- Other things equal, the larger the number of suppliers, the greater the market supply. Conversely, the smaller the number of firms in the industry, the less the market supply.

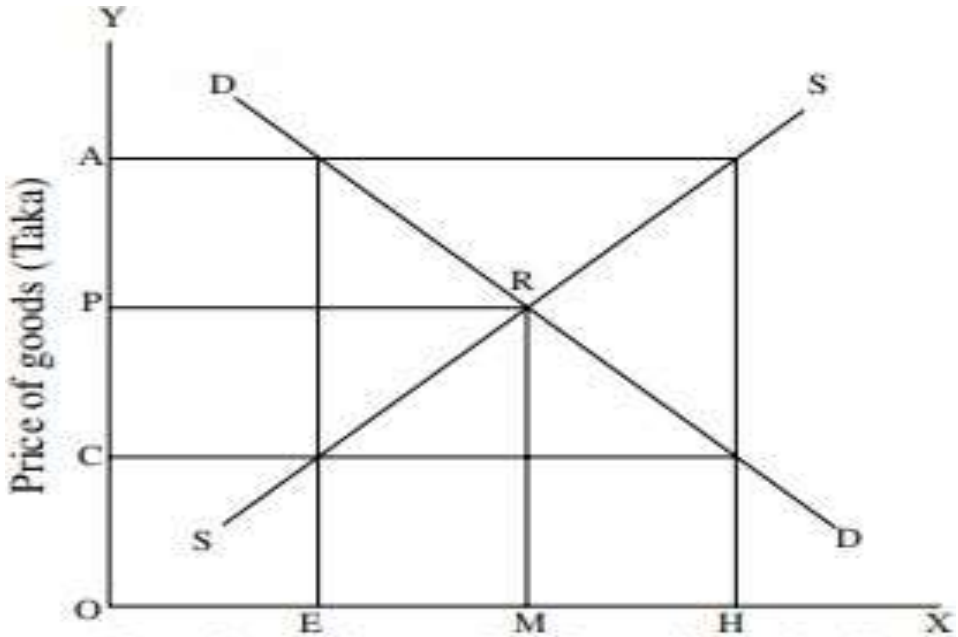
Supply and Demand Relationship

The relationship is an inverse one. When demand increases, supply decreases. When supply decreases, demand increases.



Deriving the Equilibrium Price

The price at which the demand and the supply are the same is known to be the equilibrium price. At the equilibrium price the quantity of the good that is bought or sold is known as the equilibrium quantity. An ordinary picture of the market is that the bargain between the buyer and the seller for the price of the good. The buyer tries to buy the good at the lowest possible price. On the other hand the seller tries to sell it at the highest possible price. Through this bargain between the buyer and the seller, ultimately the good is bought or sold at a price where its demand and the supply are the same.



Deriving the Equilibrium Price

OP is the equilibrium price and the equilibrium quantity is OM. The lines DD and SS intersect at the point R. This intersecting point R indicates the equilibrium price and the equilibrium quantity.

Deriving the Equilibrium Price

Suppose, at OA price, the quantity of the demand and the supply are respectively OE and OH. In this price, the quantity of supply is greater than the quantity of demand. When the supply is greater than the demand, a downwards pressure is created on the price. Thus, the price will fall from OA to OP. Here the demand and the supply will be the same. And if the price is OC, then the quantity of supply is OE and quantity of demand is OH. That is, here the demand is greater than the supply and therefore, the price of the good will definitely rise and the price will get stable at OP. There also, the demand and the supply are the same. Thus it can be seen that when the price is OP, only then the total quantity of the demand and the supply are the same that is OM. That means, at price OP, demand and supply are the same and there is no tendency of any rise or fall in the price.

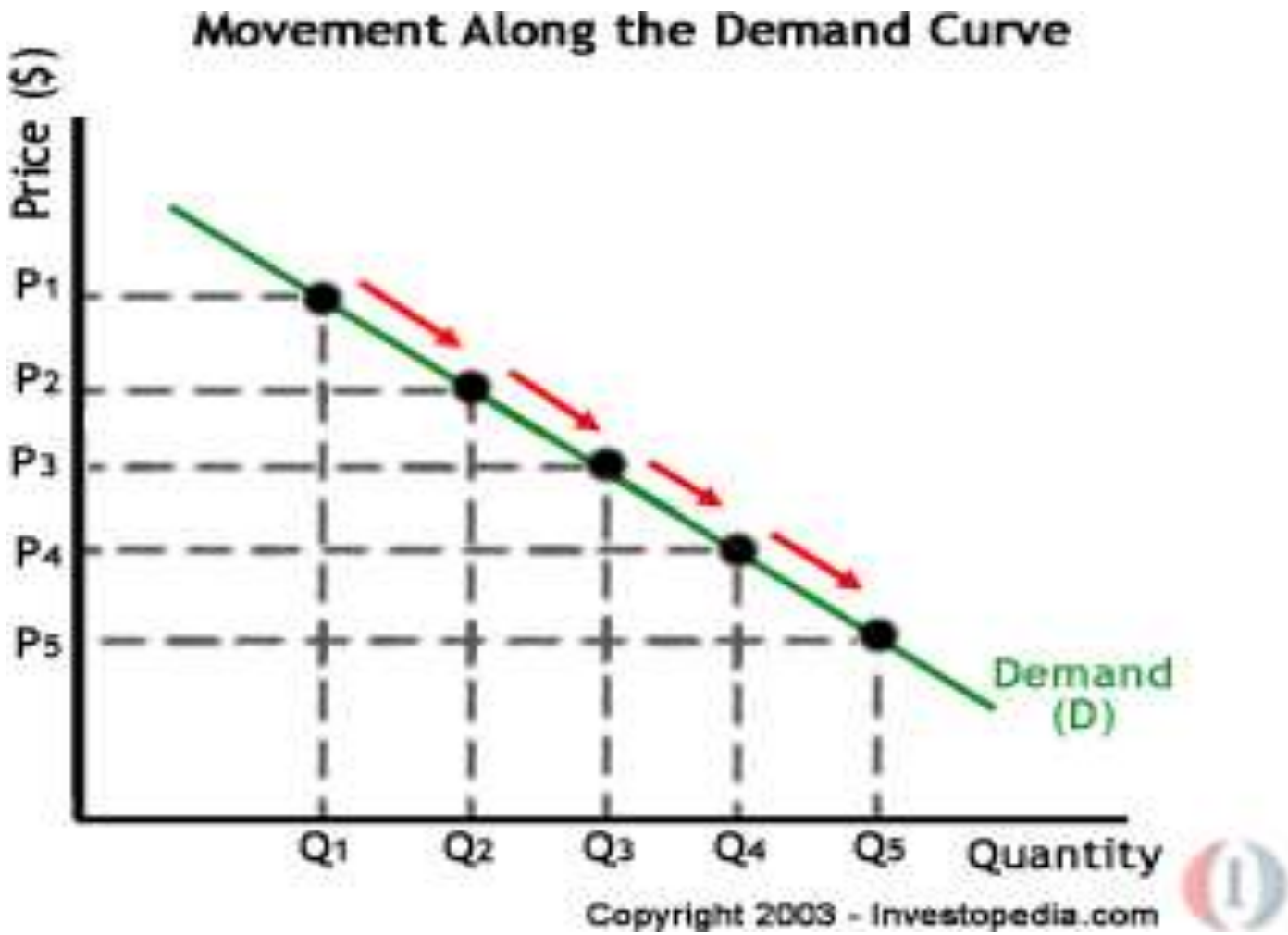
In the real market place equilibrium can only ever be reached in theory, so the prices of goods and services are constantly changing in relation to fluctuations in demand and supply.

Shifts vs. Movement

1. <u>Movements</u>

A movement refers to a change along a curve. On the demand curve, a movement denotes a change in both price and quantity demanded from one point to another on the curve.

The movement implies that the demand relationship remains consistent. Therefore, a movement along the demand curve will occur when the price of the good changes and the quantity demanded changes in accordance to the original demand relationship. In other words, a movement occurs when a change in the quantity demanded is caused only by a change in price, and vice versa.


Movement Along the Demand Curve
Price ($)
P1
P2
P3
P4
P5
Demand
(D)
Q1
Q2
Q3
Q4
Q5
Quantity
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Like a **movement** along the demand curve, a movement along the supply curve means that the supply relationship remains consistent.

Therefore, a movement along the supply curve will occur when the price of the good changes and the quantity supplied changes in accordance to the original supply relationship. In other words, a movement occurs when a change in quantity supplied is caused only by a change in price, and vice versa.


Movement Along the Supply Curve
Price ($)
P1
P2
P3
P4
P5
Supply
(S)
Q5
Q4
Q3
Q2
Q1
Quantity
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2. Shifts

A shift in a demand or supply curve occurs when a good's quantity demanded or supplied changes even though price remains the same.

For instance, if the price for a bottle of beer was $2 and the quantity of beer demanded increased from Q1 to Q2, then there would be a shift in the demand for beer.

Shifts in the demand curve imply that the original demand relationship has changed, meaning that quantity demand is affected by a factor other than price. A shift in the demand relationship would occur if, for instance, beer suddenly became the only type of alcohol available for consumption


Shift in Supply for Beer
Price ($)
S2
S1
2
Q2
Q1
Quantity
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Conversely, if the price for a bottle of beer was $2 and the quantity supplied decreased from Q1 to Q2, then there would be a shift in the supply of beer.

Like a shift in the demand curve, a shift in the supply curve implies that the original supply curve has changed, meaning that the quantity supplied is effected by a factor other than price. A shift in the supply curve would occur if, for instance, a natural disaster caused a mass shortage of hops; beer manufacturers would be forced to supply less beer for the same price.